Table 1: Per-object contact fidelity and dynamic quality on CORE4D-G1 (all 4 methods share the same 52 sequences / 11 objects). ↑: higher is better, ↓: lower is better. **Bold** = best, underline = second best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours, SBTO and OmniRetarget are computed on a MuJoCo physics rollout (native for the two optimizers; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable; GMR's are from its own kinematic pipeline at a different scale and are indicative only (read jointly with contact).

		Phys. contact@3mm ↑				Phys. contact@10mm ↑				Phys. pen@3mm ↓				Phys. pen@10mm ↓				Max phys pen (mm) ↓				Geom. pen@2mm ↓				Ankle jerk p95 (m/s^3) ↓				Obj speed max (m/s) ↓				Foot slip max (m) ↓			
Object	n	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR	Ours	SBTO	OmniRt	GMR
box001	6	**0.612**	0.234	0.015	0.001	**0.821**	0.560	0.289	0.013	**0.168**	0.343	0.607	0.761	**0.010**	0.077	0.419	0.748	**9.1**	16.5	17.2	220.7	**0.038**	0.208	0.507	0.761	540.1	854.9	1315.3	**136.8**	**1.14**	1.49	1.95	1.29	**0.83**	1.21	1.23	0.85
box004	3	**0.486**	0.167	0.064	0.006	0.679	0.608	**0.689**	0.009	**0.123**	0.431	0.571	0.525	**0.003**	0.146	0.171	0.522	**7.2**	38.0	13.2	128.3	**0.054**	0.421	0.541	0.528	805.2	1136.9	1555.5	**184.1**	**1.36**	1.89	3.51	2.27	1.10	1.19	1.13	**0.77**
box021	11	**0.552**	0.172	0.149	0.006	0.761	0.639	**0.904**	0.018	**0.180**	0.522	0.579	0.599	0.019	0.151	**0.001**	0.587	14.3	36.0	**8.6**	165.7	**0.062**	0.473	0.225	0.601	707.7	1184.4	1638.7	**176.6**	**1.32**	1.79	3.54	2.27	1.10	1.45	1.38	**0.79**
box023	7	**0.583**	0.194	0.281	0.003	0.731	0.588	**0.774**	0.021	**0.110**	0.327	0.337	0.322	**0.017**	0.078	0.039	0.304	20.9	21.6	**10.6**	92.0	**0.076**	0.327	0.272	0.324	739.9	1039.1	1743.0	**212.6**	**1.52**	2.18	3.05	2.04	0.79	1.07	1.25	**0.57**
box024	4	**0.510**	0.289	0.000	0.003	**0.860**	0.821	0.104	0.003	**0.264**	0.480	0.751	0.347	**0.004**	0.052	0.662	0.347	**8.3**	14.1	16.9	114.8	**0.037**	0.194	0.526	0.347	613.8	1132.4	1611.8	**152.0**	**1.40**	1.99	2.28	1.48	1.04	1.53	1.53	**0.93**
bucket003	3	**0.564**	0.259	0.009	0.004	**0.751**	0.701	0.105	0.020	0.250	0.452	0.544	**0.227**	**0.109**	0.120	0.462	0.210	**15.8**	20.8	33.7	50.8	**0.038**	0.433	0.378	0.453	665.8	1092.0	1547.2	**178.9**	**1.36**	1.73	2.12	1.41	**0.65**	0.86	1.52	1.51
bucket007	4	**0.492**	0.156	0.062	0.012	**0.798**	0.531	0.240	0.022	0.202	0.552	0.522	**0.139**	**0.013**	0.270	0.414	0.129	**11.2**	29.4	66.9	44.8	**0.055**	0.542	0.514	0.334	691.5	884.3	1324.3	**167.6**	**1.17**	1.64	2.32	1.55	**0.60**	1.02	1.14	1.27
chair006	4	**0.711**	0.145	0.038	0.000	**0.925**	0.762	0.243	0.000	0.173	0.633	0.472	**0.000**	0.002	0.139	0.344	**0.000**	7.4	18.5	29.8	**0.0**	**0.045**	0.536	0.367	0.668	482.1	851.7	1437.1	**135.5**	**1.08**	1.45	1.85	1.22	**0.92**	1.14	1.16	1.21
desk007	3	**0.735**	0.263	0.014	0.000	**0.862**	0.785	0.225	0.000	0.154	0.596	0.842	**0.000**	0.023	0.098	0.669	**0.000**	18.2	24.3	45.9	**0.0**	**0.062**	0.670	0.814	0.152	527.0	733.0	1181.6	**106.9**	**1.06**	1.34	1.56	1.07	**0.52**	1.19	1.16	1.39
desk021	4	**0.619**	0.100	0.149	0.006	**0.857**	0.577	0.627	0.026	0.184	0.654	0.346	**0.089**	**0.002**	0.268	0.008	0.069	**7.2**	24.7	10.7	40.1	**0.026**	0.553	0.180	0.198	704.7	1296.9	1716.8	**159.1**	**1.28**	1.82	4.28	2.87	**0.89**	1.11	1.79	1.43
desk023	3	**0.685**	0.136	0.101	0.006	**0.921**	0.747	0.506	0.008	0.170	0.553	0.421	**0.068**	**0.000**	0.135	0.144	0.066	**6.4**	20.3	17.2	60.6	**0.044**	0.489	0.285	0.226	610.5	1064.4	1943.9	**246.8**	**1.31**	1.89	3.59	2.36	0.65	0.93	1.36	**0.64**
Overall	52	**0.588**	0.190	0.101	0.004	**0.804**	0.650	0.510	0.014	**0.176**	0.490	0.536	0.349	**0.017**	0.136	0.247	0.339	**12.3**	25.1	21.1	102.1	**0.052**	0.425	0.381	0.456	653.8	1041.7	1560.8	**170.4**	**1.29**	1.77	2.83	1.86	**0.87**	1.20	1.33	0.96